▣ 02 Probability rules and constructs

Remember that Roman letters (A, B, C) denote events. Also, we use the terminology Pr(A) to denote the probability of event A.

De Morgan's laws

$\overline{(A \cup B)} = \overline{A} \cap \overline{B}$ $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$ $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

The correctness of these rules can be seen by sketching Venn diagrams….
Or by considering the logic of an applied example.

Example

Let A be the event that the integrity of an important data set on a computer's hard drive is retained.
Let B be the event that integrity of a backup is retained on a memory stick.
An event of obvious interest is that the data is retained in some form which is represented by $A \cup B$.
For this event not to occur, both A and B must not occur.
Thus $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$.

The left-hand side might be read – *The (complete) data is neither on the hard drive nor the memory stick*. The right-hand side might read – *The data is not on the hard drive, and it is not on the memory stick.*

Problem 1

Demonstrate, using a Venn diagram that $D = (A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

As an example, define the following events.
A: I have an electric, cordless drill. B: I have a charged battery for the cordless drill
C: I have a corded drill. D: I can drill a hole.

Exercise

Demonstrate, using a Venn diagram that $D = (A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ and write down an interpretation of each version of D for the events defined below

A: My smartphone is operational B: My landline is operational
C: The pizza shop is open for deliveries D: I can order a pizza

Probability range

For any event E, let p = Pr(E), then $0 \le p \le 1$.
If p = 0 then the event is almost surely impossible, if p = 1 then the event almost surely must occur.

Curiously an event could have probability 0 and still occur (e.g., the event of flipping a fair coin endlessly and never getting tails).
Similarly, an event could have probability 1 and never occur (e.g., the event of flipping a fair coin endlessly and eventually getting tails)

Complement law

For any event E, $Pr(E) + Pr(\neg E) = 1$ or $Pr(\neg E) = 1 - Pr(E)$

For example, if E is the event of getting at least one club in a hand of 5 cards, then ¬E is the event of not getting at least one club which is the same as getting no clubs.
Often it will be easier to calculate $Pr(\overline{E})$ than Pr(E) as we will see in some later examples.

Additional law

$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$

Example

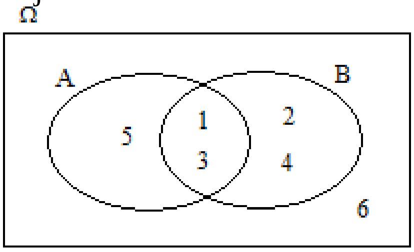
Consider the single die roll experiment again.
A : outcome is odd (i.e., one of 1, 3 or 5)
B : outcome is at most 4 (i.e., 1, 2, 3 or 4)

Note that $A \cap B$ is the event that the outcome is both odd <u>and</u> at most 4
And $A \cup B$ is the event that the outcome is odd <u>or</u> at most 4 <u>or</u> both

<u>Problem 2</u>

Calculate Pr(A), Pr(B), $\Pr(A \cap B)$ and $\Pr(A \cup B)$

A justification for the addition law can be made by referring to the Venn diagram above.



Note that $\Pr(A \cup B) = \Pr(A) + \Pr(B) \Leftrightarrow$ A and B are disjoint events (<u>Simple addition law</u>)

Law of total probability

Example

For a randomly selected member of some population define the events C, E and N to be that the subject is a current smoker, an ex-smoker and a never smoker respectively.

Note that these events are mutually exclusive and collectively exhaustive.

Let A be the event that the subject suffers from asthma and suppose that Pr(A) is required.

Then $\Pr(A) = \Pr(A \cap C) + \Pr(A \cap E) + \Pr(A \cap N)$

This is an example of <u>the total law of probability</u>.

Example

Consider a statistics examination taken by 200 students from the schools of engineering, science and business. Assume the results are as tabulated.

	Engineering	Science	Business
Pass	48	49	63
Fail	12	21	7

If one of the 200 students are randomly selected, how likely it is that they will have passed the exam?

Solution 1:
Total number of that pass = 48 + 49 + 63 = 160 = n. Total of all students = 200 = N.
So, the required likelihood or probability is = $^{n}/_{N}$ = $^{160}/_{200}$ = 0.8 or 80%

<u>Problem 3</u>

Show that the same result of 0.8 is obtained using the total law of probability.

Note that for any event B, the pair of events B and ¬ B are mutually exclusive and collective exhaustive.
So, for any events A and B, Pr(A) = Pr(A ∩ B) + Pr(A ∩ ¬ B)

Example
If 11% of a population are unemployed males and 8% are unemployed females, then clearly the overall unemployment rate is 11% + 8% = 19%.

<u>Problem 4</u>

Show that result above holds using probability notation.
I.e., let G: person is unemployed and let M: person is male

Conditional probability

Sometimes it is necessary to adjust a measure or probability in light of additional knowledge about our experiment.

For example, a public health official might warn that the prevalence of a particular illness is 3% in the general population, but this increases to 8% amongst smokers.

If we let I be the event that a member of the public has this illness, then Pr(I) = 0.03.

If we let S be the event that a member of the public is a smoker then we write Pr(I | S) to mean the probability that they have this illness <u>given that</u> or <u>conditional on the fact that</u> he is a smoker.

For obvious reasons Pr(I | S) is called a conditional probability and evaluates here to 0.08.

- *Take care not to confuse PR(A| B) with Pr(B | A). I.e., the percentage of the asthma suffering population who are smokers is not the same as the percentage of the smoking population who are asthma sufferers.*

As an extreme example of how different they might be, suppose that 0.1% of a population are incarcerated (in prison) and 90% of the prison population are male.

Then Pr(Incarcerated | Male) = 0.1% but Pr(Male| Incarcerated) = 90%

Pr(A| B) and Pr(B | A) can however be related in a very important equation due to Bayes, which we see later.

Exercise: A survey of all 100 employees in a firm are asked how they most usually travel to work - the results are <u>cross tabulated</u> with their gender (opposite).

	Walk	Drive	Bus
Male	8	30	15
Female	12	15	20

Define W: employee walks to work and similarly define D and B
Define M: employee is male so ¬ M : employee is female.

Interpret each of the following events and write down their probabilities

M, W, D ∪ B, B ∪ D, D ∩ B, B | M, M| B, ¬ M | D, M | (W ∪ D)